



Beneficial effects of the herbal medicine zuo gui wan in a mice model of Alzheimer's disease via Drp1-Mediated inhibition of mitochondrial fission and activation of AMPK/PGC-1 α -regulated mitochondrial bioenergetics

Xirui Yan^{a,b}, Yifang Yang^a, Weiling Huang^a, Shuping Fu^c, Bo Cui^d, Min Chu^e, Yang Dong^e, Yinting Peng^f, Haiyan Song^a, Jianrong Shi^{a,**}, Qing Liu^{a,*}

^a School of Traditional Chinese Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai, 201203, China

^b Department of Otolaryngology, Head and Neck Surgery, Shanghai Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

^c Key Laboratory of Acupuncture and Medicine Research of Ministry of Education, Nanjing University of Chinese Medicine, Nanjing, 210023, China

^d School of Pharmacy, Shanghai University of Traditional Chinese Medicine, Shanghai, 201203, China

^e School of Integrative Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai, 201203, China

^f School of Rehabilitation Science, Shanghai University of Traditional Chinese Medicine, Shanghai, 201203, China

ARTICLE INFO

Keywords:

Zuo gui wan
Metformin
Mitochondrial
 $A\beta$
Drp1
AMPK
Alzheimer's disease

ABSTRACT

Ethnopharmacological relevance: Zuo Gui Wan (ZGW) is a well-known traditional Chinese medicine decoction used for approximately 400 years to treat age-related degenerative conditions, including cognitive impairment in older adults, osteoporosis, and general aging. However, the mechanism of action for ZGW remains unclear.

Aims of the study: This study aims to investigate the efficacy of ZGW in improving cognitive function in Alzheimer's disease (AD) animal models and to explore the underlying mechanisms, presenting a novel perspective in the field.

Materials and methods: Six-month-old male APP/PS1 mice were divided into three groups that received either metformin (200 mg/kg daily) or ZGW (6 and 12 g/kg daily). High-performance liquid chromatography was conducted for ZGW's quality control. Cognitive function was assessed using the Morris water maze test. Neuronal loss, synaptic plasticity, and β -amyloid ($A\beta$) deposition were evaluated through Western blot or immunofluorescence staining. The underlying molecular mechanisms were investigated using ELISA, Western blot, qRT-PCR, co-immunoprecipitation assay, ATP assay, and cytochrome *c* oxidase assay.

Results: ZGW, administered in both low and high doses, significantly enhanced cognitive performance, notably decreased neuronal loss and $A\beta$ deposition, and reduced levels of $A\beta$ 1-40/42. It also inhibited excessive mitochondrial division primarily by suppressing phosphorylated dynamin-related protein 1 (Drp1), especially at high doses of ZGW. Co-immunoprecipitation experiments further confirmed that ZGW inhibited the interaction between $A\beta$ and p-Drp1. Furthermore, similar to the effects of the AMP-activated Protein Kinase (AMPK) activator metformin, ZGW led to a marked increase in the mitochondrial DNA copy number and upregulated the AMPK/PGC-1 α /NRF1/TFAM pathway. Improvements in mitochondrial function were evident from the increased ATP production, elevated expression of superoxide dismutase 2, and upregulated cytochrome *c* oxidase activity. Additionally, the excess byproduct of reactive oxygen species, 4-hydroxy-2-nonenal, decreased in the group treated with ZGW.

Conclusion: This study provides compelling evidence that ZGW improves cognitive impairment in APP/PS1 mice by activating AMPK/PGC-1 α -regulated mitochondrial bioenergetics and inhibiting $A\beta$ -induced mitochondrial fragmentation, highlighting its potential as an effective therapeutic strategy for AD.

1. Introduction

Alzheimer's disease (AD) is a prevalent neurodegenerative disorder

* Corresponding author. No. 1200, Cailun Rd School of Chinese Medicine Shanghai University of Traditional Chinese Medicine Shanghai, 201203, China.

** Corresponding author. No. 1200, Cailun Rd School of Chinese Medicine Shanghai University of Traditional Chinese Medicine Shanghai, 201203, China.

E-mail addresses: sjr@shutcm.edu.cn (J. Shi), qingliu@shutcm.edu.cn (Q. Liu).

Abbreviations:		mtDNA	mitochondrial DNA
AD	Alzheimer disease	COX	cytochrome c oxidase
Aβ	β-amyloid	WT	wild-type
APP	amyloid-β protein precursor	SYP	synaptophysin
AMPK	AMP-activated Protein Kinase	BACE1	β-site APP cleaving enzyme 1
Drp1	dynamain-related protein 1	OPA1	optic atrophy-1
TFAM	mitochondrial transcription factor A	MFN2	mitofusin 2
PGC-1α	peroxisome proliferator-activated receptor gamma coactivator-1alpha	FIS1	mitochondrial fission 1 protein
ZGW	Zuo Gui Wan	NRF1	nuclear respiratory factor 1
MWM	Morris water maze	NRF2	nuclear respiratory factor 2
APP/PS1	APP ^{Swe} /PSEN1 ^{dE9}	SOD2	superoxide dismutase 2
HPLC	high-performance liquid chromatography	4-HNE	4-hydroxy-2-nonenal
		TGN	trans-Golgi network

that has emerged as a significant global health concern. It is the primary cause of cognitive decline in older individuals. In patients with AD, the mental decline tends to deteriorate progressively, and reversing its decline is challenging. Hence, it is imperative to initiate interventions at the early stage. There are extensive studies discussing the pathogenic mechanisms of AD, particularly focusing on mitochondrial abnormalities that occur during the early stage of this disease (Lin and Beal, 2006; Wang et al., 2020). For instance, in brain tissue from post-mortem AD patients, mitochondrial deficits have been identified as a significant trigger contributing to impaired synaptic function (Wang et al., 2023). In humanized Aβ-knockin mice (Jia et al., 2023) and APP/PS1 mouse model (Fang et al., 2019), mitochondrial dysfunction occurring early in AD, was found to have the potential to initiate β-amyloid (Aβ) accumulation, as well as influence Aβ and tau pathology. In contrast, the build-up of Aβ, due to the cleavage of the amyloid-beta protein precursor (APP) by specific enzymes, can disrupt mitochondrial function. (Bobba et al., 2013; Manczak et al., 2006; Zysk et al., 2023). As a result, the Aβ deposits can exacerbate mitochondrial dysfunction, contributing to a detrimental cycle.

Maintaining sufficient functional mitochondria in healthy morphology is crucial to neurons. This process involves fine regulation of mitochondrial dynamics (the hemostasis of mitochondrial fusion/fission) and mitochondrial biogenesis (generating new mitochondria). In AD, mitochondrial dynamics will be disrupted, resulting in a preference for mitochondrial fission. (Eckert et al., 2012; Islam et al., 2019). Investigating the inhibition of excessive mitochondrial fission and the restoration of mitochondrial dynamics offers a new perspective on neurodegenerative conditions, including AD (Chen et al., 2023b; Grel et al., 2023). Recent studies indicated that inhibiting dynamain-related protein 1 (Drp1) can improve mitochondrial morphology, reduce lipid peroxidation, and lower Aβ deposition in the brains of mouse models with AD (Mishra and Thakur, 2023; Wang et al., 2017). In addition to these factors, mitochondrial transcription factor A (TFAM) and peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PGC-1α) are majorly involved in the regulation of impaired mitochondrial bioenergetics in the pathogenesis of AD. Accordingly, administration of recombinant human TFAM (rhTFAM) or upregulation of PGC-1α was found to improve memory in aged mice (Thomas et al., 2012) as well as in 3X Tg AD mice (Li et al., 2022; Oka et al., 2016). This evidence highlights the critical impact that dysfunctional mitochondrial populations and morphology can exert in AD and the improvement of mitochondrial function as a prospective direction for AD treatment.

Zuo Gui Wan (ZGW), a well-known Chinese herbal formula, consists of eight Chinese medicines: Radix Rehmanniae, Fructus Lycii, Fructus Corni, Semen Cuscutae, Radix Achyranthis Bidentatae, Rhizoma Dioscoreae, Carapax et Plastrum Testudinis, and Cornu Cervi Pantotrichum. In China, ZGW has been utilized for approximately 400 years to address age-related degenerative conditions, such as cognitive impairment in

older people, osteoporosis, and general aging. (Cen et al., 2023; Li et al., 2018). Previous studies have reported that ZGW improved mitochondrial energy metabolism and mitochondrial membrane potential in aging rat models (Yao et al., 2017; Zhao et al., 2019). A recent study by Liu et al. (2022) demonstrated that ZGW enhanced cognitive performance in rats with AD induced by streptozotocin. The findings suggest that ZGW may exert its effects by suppressing neuroinflammation, which is closely associated with mitochondrial dysfunction. These studies indicate that ZGW can improve mitochondrial dysfunction during aging or in the context of AD pathology; however, the mechanism underlying these effects is unclear. Therefore, this study sought to fill in the gap in this topic through experiments on APP^{Swe}/PSEN1^{dE9} (APP/PS1) AD model mice. The results showed that ZGW ameliorated the cognitive deficits in the model mice by attenuating the interaction between Aβ and p-Drp1 while restoring the dynamic balance between mitochondrial division and fusion. In addition, ZGW was found to activate AMPK/PGC-1α-regulated mitochondrial bioenergetics. Our data provide evidence for the effect of ZGW on mitochondrial dynamics and biogenesis in AD and further indicate that enhancement of mitochondrial function via ZGW may represent a novel strategy for treating AD.

2. Materials and methods

1. Preparation of Zuogui Wan

Rehmanniae Radix Praeparata (lot# 200822), *Dioscoreae Rhizoma* (lot# 200714), *Cuscutae Semen* (lot# 200902), *Achyranthis Bidentatae Radix* (lot# 200717), *Corni Fructus* (lot# 200908), and *Fructus Lycii* (lot# 200822), were purchased from Kang-Qiao Co. Ltd. *Cervi Cornus Colla* (lot# Z37021395) and *Testudinis Carapacis et Plastri Colla* (lot#

Table 1
Data on herbal drugs utilized in this investigation.

Drug name (Pharmacop. of China, 2020)	Plant or animal of origin	Parts of Plant or animal	Ratio in the ZGW
Rehmanniae Radix Praeparata	<i>Rehmannia glutinosa</i> (Gaertn.) Libosch. ex DC.	Root	8
Dioscoreae Rhizoma	<i>Dioscorea oppositifolia</i> L.	Rhizome	4
Cuscutae Semen	<i>Cuscuta chinensis</i> Lam.	Seed	4
Achyranthis Bidentatae Radix	<i>Achyranthes bidentata</i> Blume.	Root	3
Corni Fructus	<i>Cornus officinalis</i> Siebold & Zucc.	Fruit	4
Lycii Fructus	<i>Lycium barbarum</i> L.	Fruit	4
Cervi Cornus Colla	<i>Cervus elaphus Linnaeus</i>	Horn	4
Testudinis carapacis et plastri Colla	<i>Chinemys reevesii</i>	Shell	4

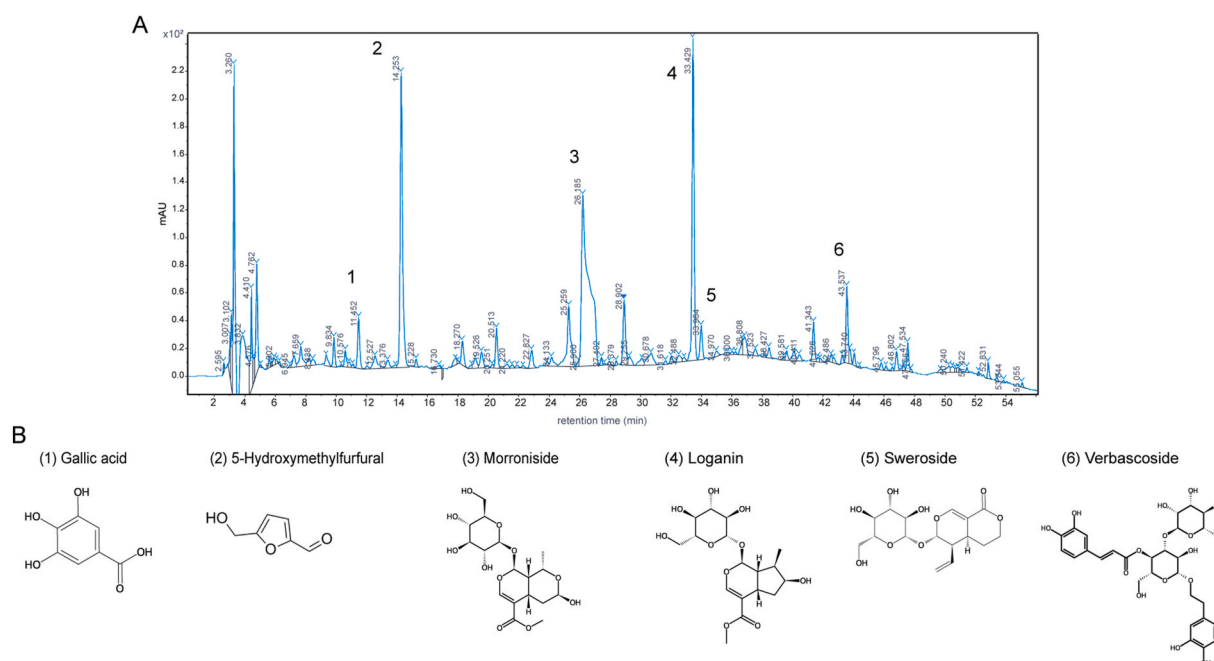


Fig. 1. HPLC chromatogram of the ZGW formulation

(A) HPLC chromatograms of ZGW detected at 238 nm. (B) The chemical structures of six compounds identified in the ZGW were obtained from the ChemSpider chemical database.

Z20043493), were purchased from Dong-Er-Jiao Co. Ltd. All drugs were prepared in an 8:4:4:3:4:4:4:4 ratio by weight (Table 1), as specified in the Chinese Medicine Formulae Images Database (<https://library.hkbu.edu.hk/electronic/libdbs/cmfid/>). All drugs were immersed in distilled water at a ratio of 1:8 for 12 h, then boiled for 60 min and filtered. The residue was boiled for another 60 min in 1 L of distilled water. As previously described, the two extracts were mixed, filtered, and lyophilized (Liu et al., 2018, 2022).

2. Quality control of ZGW by High-performance liquid chromatography (HPLC)

HPLC analysis of ZGW was conducted using the Agilent 1260 Infinity HPLC System with a variable wavelength detector. The components were separated using a Boltimate™ Ultimate® XB-C18 chromatography column. The mobile phase applied in this study is acetonitrile-water (solvent A) and 0.1% formic acid (solvent B) with a flow rate of 1 ml/min at 35 ± 1 °C. Gradient elution was performed from 2% to 10% (solvent A, 0–20 min), 10%–15% (solvent A, 20–30 min), 15%–25% (solvent A, 30–45 min), and 25%–70% (solvent A, 45–60 min). The peaks in the HPLC chromatogram of ZGW were verified to authentic standards by comparing retention times and UV spectra. Six compounds were identified: verbascoside, 5-hydroxymethylfurfural, gallic acid, loganin, morroniside, and sweroside (Fig. 1). Table 2 shows each compound's retention times and peak areas as measured at 238 nm wavelength.

Table 2

Reference compounds identified in ZGW by HPLC analysis.

Compounds	Formula	RT (min)	Peak area
Gallic acid	C ₇ H ₆ O ₅	11.452	343.5
5-Hydroxymethylfurfural	C ₆ H ₆ O ₃	14.253	2320.5
Morroniside	C ₁₇ H ₂₆ O ₁₁	26.185	4185
Loganin	C ₁₇ H ₂₆ O ₁₀	33.429	2165.9
Sweroside	C ₁₆ H ₂₂ O ₉	33.964	258.5
Verbascoside	C ₂₉ H ₃₆ O ₁₅	44.02	72.5

3. Animals and experimental treatments

Six-month-old male APP/PS1 transgenic mice and wild-type (WT) mice were purchased from Shanghai Model Organisms Center Inc. Male APP/PS1 mice were randomly divided into four groups containing 12 mice each: (1) untreated APP/PS1 mice, (2) APP/PS1 mice treated with metformin (200 mg/kg daily) (Farr et al., 2019), (3) APP/PS1 mice treated with ZGW at a dosage of 6 g/kg daily that was orally administered as part of their chow, and (4) APP/PS1 mice treated with ZGW at a higher dosage of 12 g/kg daily orally administered in chow. Our estimate for the ZGW dosage was based on a mouse weighing 35 g, which has an average chow consumption of 5 g per day, according to our pilot study, which was also consistent with a previous study by Niedowicz DM et al. (Niedowicz et al., 2018). To ensure the accurate dosage of ZGW for each mouse, we recorded the feed consumption and monitored their body weight as described by Yakushiji Y et al. (Yakushiji et al., 2020). Based on clinical experience and clinical studies, a person weighing 60 kg was administered ZGW at a dosage ranging from 27 to 36 g per day (Deng, 2017; Li et al., 2018). To convert this human dosage into the animal equivalent dose (AED) based on body surface area, we divide the human dose by 0.081 (Nair and Jacob, 2016). This calculation yields an equivalent mouse dose of 6 g/kg/day for the low dosage, while the high dosage is twice the AED. Normal mouse chow and chow containing ZGW were prepared by Shanghai FBSH Biopharmaceutical Co. Ltd. All procedures were approved by the Animal Ethics Committee of Shanghai University of Traditional Chinese Medicine (no. PZSHUTCM210122001).

4. Morris water maze test

The maze and platform were set up as previously described (Yu et al., 2023). The MWM test includes an acquisition trial and a probe trial. In the 5-day training period of the acquisition trial, four 60-s trials were carried out per day: in each trial, the starting point differed within the quadrants, and the mice were required to locate the platform. On day 6, animals were allowed to swim for 60s without a platform (probe trial). Escape latency, swim distance, and speed in the acquisition trial or the

crossing times, time spent in the target, and the movement path in the probe test were recorded and visualized using the video-tracking system EthoVisionXT version 11.5 (Noldus, Netherlands).

5. Immunoblotting and co-immunoprecipitation experiments

Total proteins from the hippocampus were extracted using RIPA lysis buffer, which has been added to inhibitor cocktails of phosphatase and phosphatase. The measurement of protein concentration and the immunoblotting experiments were followed by the previous publication (Li et al., 2023). The membranes were incubated with primary antibodies (listed in [Supplementary Table S1](#)), and subsequently incubated with the corresponding secondary antibodies. After adding chemiluminescence substrate (ThermoFisher, USA), blots were immediately captured with the ChemiDOC™ (Bio-Rad, USA) and quantified by ImageJ.

For the co-immunoprecipitation experiments, 5 µg of p-Drp1 antibody or normal rabbit IgG were pre-incubated with magnetic beads (Beyotime, China) to induce the formation of a magnetic bead-antibody complex. Proteins extracted from three hippocampal samples (obtained from three mice) in each group were incubated with the magnetic bead-antibody complex overnight at 4 °C. The proteins were then immunoblotted with the antibodies against 6E10 and p-Drp1.

6. Quantitative RT-PCR analysis

To investigate the transcription of major regulatory genes associated with mitochondrial biogenesis, hippocampal RNA was isolated using the RNA extraction kit (Beyotime, China). Next, the cDNA synthesized from the RNA was further applied for the PCR using the SYBR qPCR kit (Beyotime, China). Primer sequences for all PCR experiments were reported in [Supplementary Table S2](#). The relative expression of genes was calculated by the $2^{-\Delta\Delta C_t}$ method and expressed as a percentage of the corresponding genes from WT animals.

7. DNA isolation and mitochondrial DNA (mtDNA) quantification

Total DNA was isolated from hippocampus tissue using a DNA extraction kit (Beyotime, China). Mitochondrial DNA copy number (mtDNA-CN) was assessed by quantifying the relative amount of mitochondrial-encoded COX1 and nuclear gene 18sRNA, which were determined by quantitative PCR. The primer sequences are listed in [Supplementary Table S2](#). The mtDNA-CN was calculated using the following calculation: $\Delta C_t = C_{tCOX1} - C_{t18s RNA}$, whereas $\Delta(\Delta C_t) = \Delta C_t$ sample – ΔC_t control.

8. Immunofluorescence staining

After gradient dehydration, mouse brain tissue was embedded and cut into 40-µm slices using a cryo-microtome (Microm, Germany). After permeabilization, the slices were incubated overnight with primary antibodies at 4 °C. Following thorough washes with PBS, slides were incubated with fluorochrome-conjugated secondary antibodies for 2 h at room temperature. For thioflavin S staining, the brain slides were incubated in 1% aqueous thioflavin solution (Sigma-Aldrich, Germany) for 10 min and then dehydrated with a graded alcohol solution. The slides were next subjected to a standard immunohistochemical protocol and examined using a confocal microscope (LSM880; ZEISS, Germany). The proportion of Aβ-positive area (%), as covered by thioflavin S staining, was calculated in relation to the total area of the analyzed region using ImageJ.

9. ATP assay

ATP levels in the hippocampus were detected using a bioluminescence technique with an ATP assay kit (Beyotime, China). Briefly, the

hippocampal tissue was lysed. The protein fraction (100 µl) was added to 100 µl ATP assay solution and mixed. Following a 5-min incubation, ATP luminescence was measured using VICTOR Nivo™ multimode plate reader (PerkinElmer, USA).

10. Cytochrome c oxidase (COX) activity assay

Using a mitochondrial isolation kit from Abcam (USA), mitochondria were extracted from the hippocampus using the centrifugation method, which involved spin at 1000 g for 10 min, followed by a second centrifugation at 12,000 g for 15 min. COX activity was measured by using a CYTOCOX1 kit (Sigma-Aldrich, USA). Briefly, 5 µL ferric cytochrome c substrate solution was added to samples to initiate the reaction. The changes in absorbance of cytochrome c were monitored at a wavelength of 550 nm. Finally, COX activity was calculated based on the normalized protein concentration.

11. Enzyme-linked immunosorbent assay

Soluble Aβ₁₋₄₂ and Aβ₁₋₄₀ in hippocampus extracts were quantified with a sandwich antibody ELISA kit (Thermo Fisher, USA). The fresh mouse hippocampal tissue was homogenized by adding ice-cold PBS containing 1% PMSF and centrifuged for 5 min at 5000 g. After measurement of protein concentration, the hippocampal homogenates were processed into a precoated, flat-bottom 96-well plate and read on a multi-mode microplate reader (PerkinElmer, Waltham, USA) at 450 nm. The concentrations of Aβ₁₋₄₂ and Aβ₁₋₄₀ were determined by fitting to the standard curve using a four-parameter algorithm.

12 Statistical analysis

All data are presented as mean ± SEM. One-way ANOVA followed by Tukey's post-hoc test was applied to indicate the statistical significance among groups. p values < 0.05 were considered a significant difference. #p < 0.05 vs. WT, ##p < 0.01 vs. WT, ###p < 0.001 vs. WT; *p < 0.05 vs. APP/PS1, **p < 0.01 vs. APP/PS1, ***p < 0.001 vs. APP/PS1.

3. Results

1. HPLC analysis of ZGW

We first prepared the ZGW aqueous extract and assessed its quality by HPLC analysis. The peaks in the HPLC chromatogram of ZGW were verified by comparing retention times and UV spectra with authentic standards ([Fig. 1](#)). The following six compounds in ZGW were identified: verbascoside, 5-hydroxymethylfurfural, gallic acid, loganin, morroniside, and sweroside. [Table 2](#) shows the retention times and peak areas for each compound recorded at a wavelength of 238 nm. The results of regression analyses and the calculated correlation coefficients are listed in [Supplementary Table S3](#).

2. Ameliorative effects of ZGW on cognitive deficits

To verify the efficacy of ZGW on improving cognitive function in AD animals, ZGW (6 or 12 g/kg daily) was administered to APP/PS1 mice for 16 weeks. Mice receiving oral administration of metformin (200 mg/kg daily) were considered as positive controls ([Fig. 2A](#)). Body weight measurements revealed no notable differences between groups throughout the study ([Fig. 2B](#)). After treatment, the MWM test was conducted to assess the spatial learning and memory ability. During the acquisition trial, the AD model group showed a significantly longer escape latency than the control group (p < 0.01) on the fourth training day. This indicated that the APP/PS1 mice took more time to locate the platform compared to the control group. On day 5, the WT mice and ZGW- or metformin-treated mice required less time to locate the platform than the APP/PS1 mice ([Fig. 2C](#)). To eliminate the possibility that

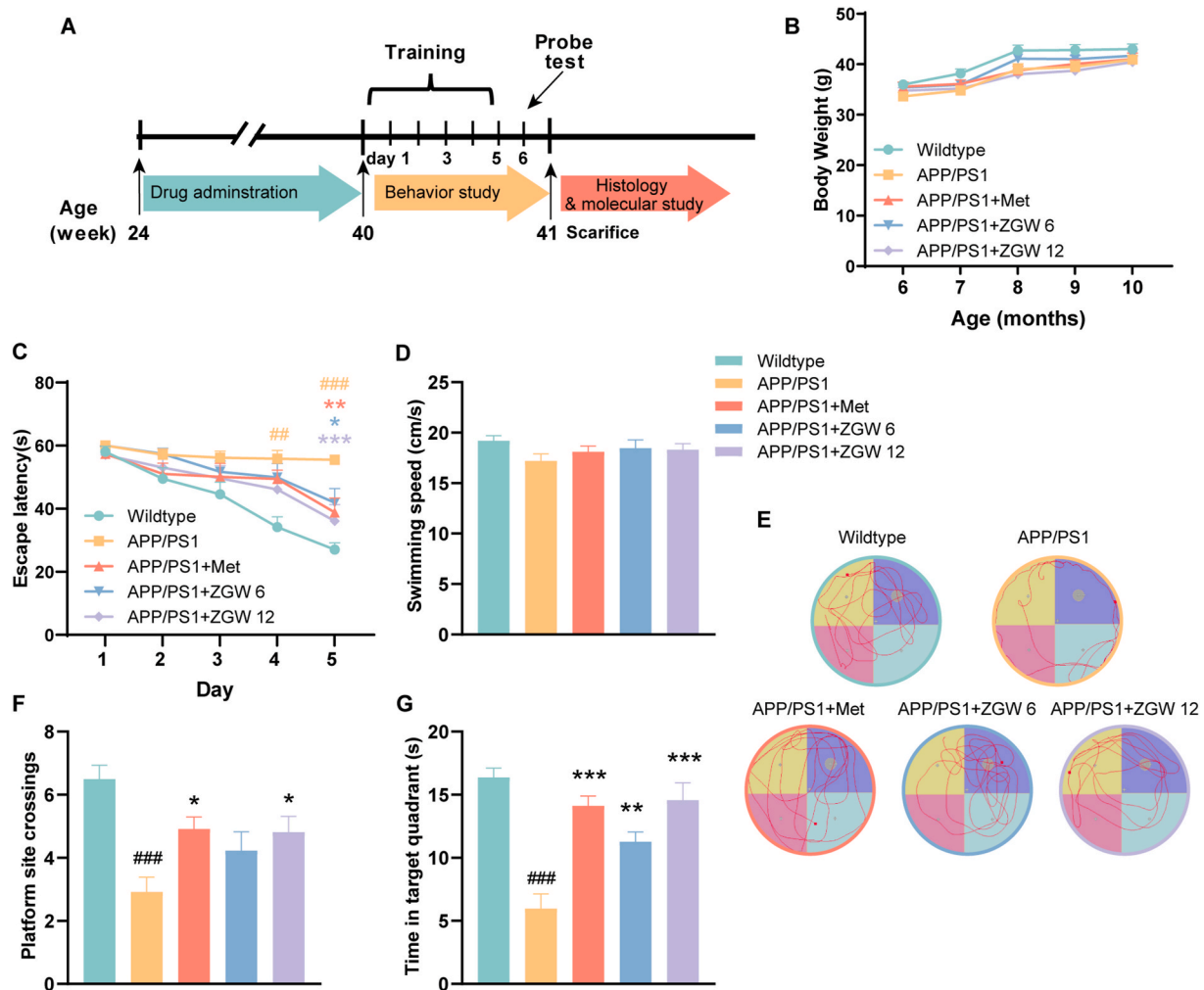


Fig. 2. Improvement of cognitive deficits in APP/PS1 mice treated with ZGW.

(A) Flow diagram of the experiment: 6-month-old male APP/PS1 transgenic mice ($n = 12$) were treated daily with either 6 or 12 g/kg ZGW, or with 200 mg/kg metformin. Behavioral experiments were conducted when animal reached 10-month-old. (B) Body weight was recorded monthly. Escape latency (C) and swimming speed (D) were recorded during the spatial acquisition phase of the MWM test. Representative motion track (E), number of crossing-over platforms (F), and time in the target quadrant (G) were recorded during the probe test.

differences in swimming ability affected latency, we analyzed swimming speed as well and found no significant variation in speed among the groups (Fig. 2D). The probe trial performed on day 6 showed that the ZGW- and metformin-treated mice exhibited a stronger tendency to search for the platform in the target quadrant compared to the APP/PS1 group. The representative swimming path of mice in each group are shown in Fig. 2E. During the probe trial, results indicated that APP/PS1 mice treated daily with ZGW (12 g/kg) or metformin crossed the previous platform location more frequently than the vehicle-treated APP/PS1 mice, suggesting improved spatial memory retention. Administration of 6 g/kg and 12 g/kg ZGW significantly increased the time spent in the target quadrant. ($p < 0.01$). Moreover, mice given the high dosage of ZGW exhibited similar performance as mice treated with metformin ($p > 0.05$) (Fig. 2G). Collectively, these data suggested that treatment with ZGW could ameliorate cognitive deficits in APP/PS1 mice.

3. ZGW decreased the loss of hippocampal neurons and improved synaptic plasticity in APP/PS1 mice.

Neuronal nuclei and filaments were labeled with NeuN and MAP2 to observe the damage to hippocampal neurons in each group (Fig. 3A). In the hippocampal CA1, CA3, and DG regions of WT mice, the neuronal cells were intact and exhibited regular morphology in a dense and neat

arrangement. APP/PS1 mice, however, showed significant neuron loss, especially in the DG and CA1 regions. It was observed that high dose ZGW or metformin significantly increased the number of neurons in these regions ($p < 0.01$) but had a weaker protective effect on neuron loss in the CA3 region ($p > 0.05$) (Fig. 3B–D). Western blot analysis showed that both ZGW and metformin increased the expression of the synapse-associated proteins CtBP2, PSD95, and synaptophysin (SYN). Notably, the high dosage of ZGW exhibited a more pronounced enhancing effect compared to metformin (Fig. 3E–H).

4. ZGW decreased the deposition of A β in the hippocampus of APP/PS1 mice

To investigate whether ZGW could reduce the deposition of A β plaques, brain tissues extracted from mice from various groups were sectioned and stained with Thioflavin S, which is a fluorescent histochemical marker of A β plaques. Brain tissue samples from APP/PS1 mice exhibited extensive A β deposition in the hippocampal region. The plaque area decreased from approximately 0.84%–0.13% and 0.15% following treatment with metformin and ZGW (12 g/kg), respectively (Fig. 4A). Furthermore, the expression of hippocampal A β_{1-42} and A β_{1-40} was significantly higher in APP/PS1 mice compared to WT mice ($p < 0.001$), while both metformin and ZGW significantly reduced

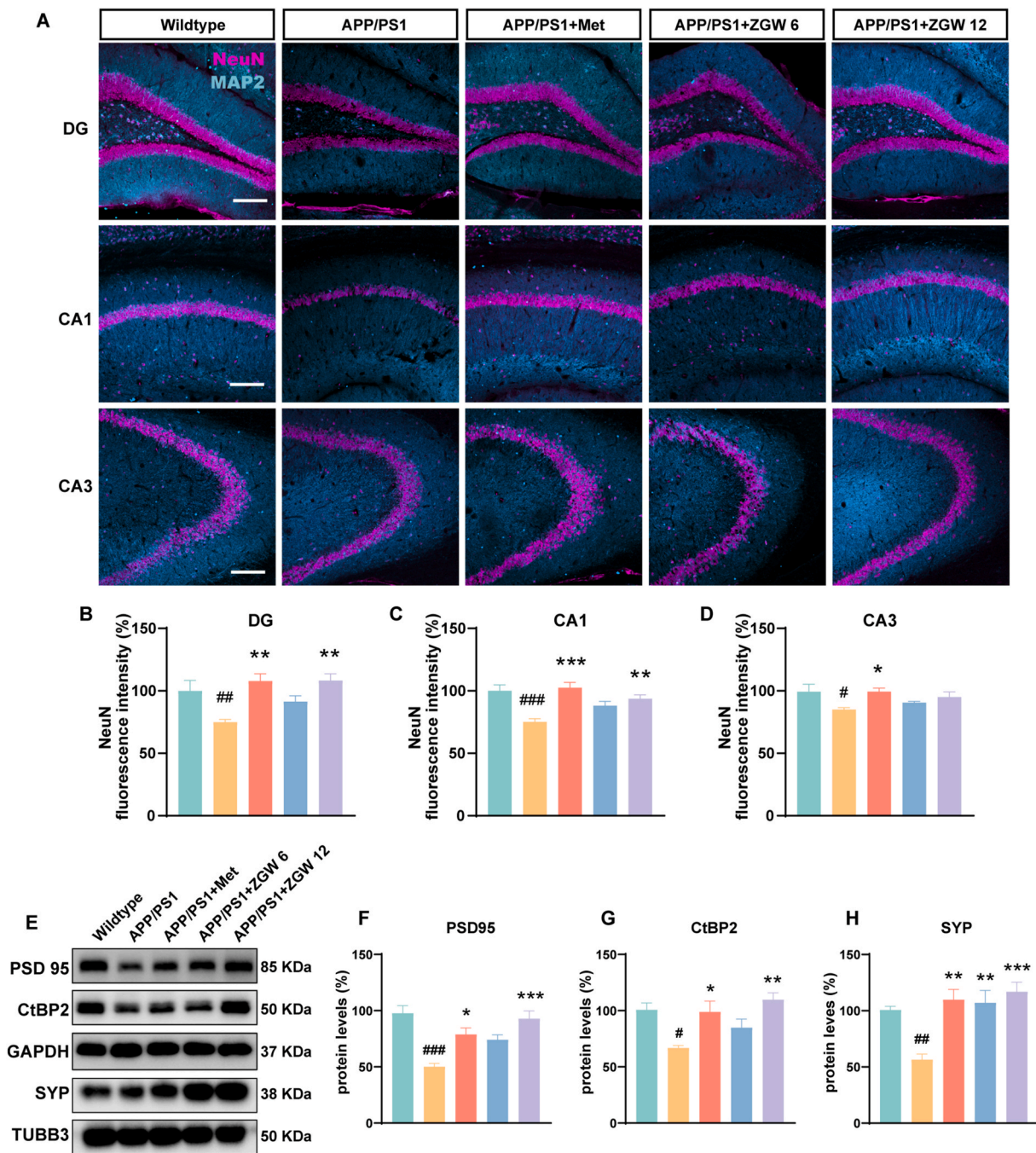


Fig. 3. ZGW decreased the loss of hippocampal neurons and improved synaptic plasticity

(A) Representative double-stained hippocampal sections illustrating co-localization of NeuN (Neuronal nuclei) and MAP2 (neurofilaments) (scale bar: 200 μ m). NeuN fluorescence intensity in the DG (B), CA1 (C), and CA3 (D) regions was quantified by ImageJ. (E) Immunoblots of PSD95, CtBP2, and SYP (synaptophysin) for hippocampal protein samples. (F–H) Statistical histograms of the results from panel A are shown.

hippocampal A β ₁₋₄₂ and A β ₁₋₄₀ level in APP/PS1 group (Fig. 4B and C). Next, we conduct Western blot to investigate the effect of ZGW on hippocampal APP and its cleaving enzyme β -site APP cleaving enzyme 1 (BACE1) (Fig. 4D). We found that APP, p-APP, and BACE1 protein expression in APP/PS1 mice was significantly higher than in WT mice (Fig. 4E–G). Compared with the APP/PS1 group, mice receiving high-dose ZGW displayed a significant reduction in p-APP expression by 41% (Fig. 4F) and a 46% decrease in BACE1 expression (Fig. 4G).

5. Inhibition of mitochondrial fragmentation by ZGW via reducing the interaction between A β and p-Drp1.

Since the deposition and accumulation of amyloid are linked to changes in mitochondrial dynamics, we further assessed the levels of proteins involved in mitochondrial fusion and fission using Western blot analysis. In comparison to the WT group, the APP/PS1 group exhibited a decrease in the expression of the optic atrophy-1 (OPA1) and Mitofusin 2 (MFN2) ($p < 0.05$). Neither metformin administration nor low-dose ZGW impacted the protein level of MFN2 and OPA1 (Fig. 5A–C). With regard to fission proteins, metformin treatment reduced p-Drp1 levels in the hippocampus of APP/PS1 mice by 56.7%, while low-dose and high-dose ZGW reduced p-Drp1 levels by 22.4% and 42.1%, respectively (Fig. 5E). The mitochondrial fission 1 protein (FIS1) expression in the

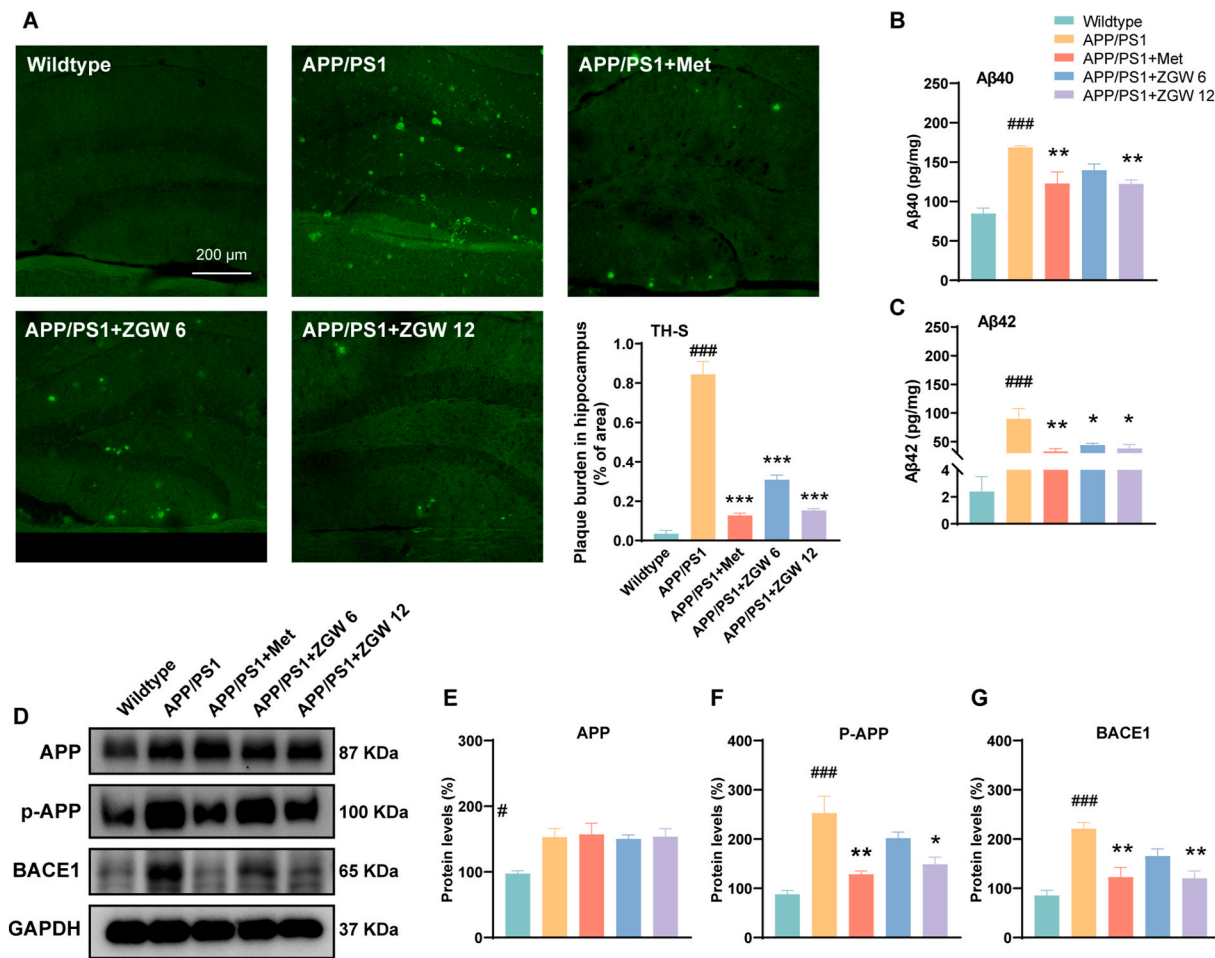


Fig. 4. ZGW decreased the deposition of A β in the hippocampus

(A) Representative images of amyloid plaques stained with Thioflavin S (green) in brain tissue sample after ZGW or metformin treatment. Amyloid plaque burden in the hippocampus was quantified using Thioflavin S fluorescence intensity analyzed by ImageJ ($n = 5$). Levels of hippocampal A β ₁₋₄₀ (B) and A β ₁₋₄₂ (C) were measured by ELISA. (D) Immunoblotting of APP, p-APP, and BACE1 in the hippocampus. (E–G) Densitometric analysis of the proteins as shown in (D) ($n = 4$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

APP/PS1 group was about 1.8 times higher than in the WT group (Fig. 5F). The treatment with metformin and ZGW (low- and high-dose) decreased the expression of FIS1. In fact, the protein level of FIS1 and p-Drp1 in the high-dose ZGW treatment group approached WT levels, indicating that ZGW may restore mitochondrial dynamics. We next examined if ZGW could reduce the interaction between A β and p-Drp1 by using a Co-IP assay (Fig. 5G). The results showed the presence of an A β peptide segment (6E10) in the hippocampus of APP/PS1 mice and the existence of an interacting force between 6E10 and p-Drp1; further, ZGW administration was found to reduce the degree of interaction between 6E10 and p-Drp1.

6. Promotion of mitochondrial biogenesis by ZGW via regulation of the AMPK/PGC-1 α pathway in the hippocampus

mtDNA is a reflection of mitochondrial biogenesis, and its replication and transcription are regulated by TFAM signaling that is stimulated by the transcription factors nuclear respiratory factor 1 (NRF1) or nuclear respiratory factor 2 (NRF2) (Fu et al., 2019). The APP/PS1 group exhibited a 58% reduction in mtDNA-CN compared to the WT group. In contrast, the metformin, low-dose ZGW, and high-dose ZGW groups showed significant increases in mtDNA-CN, which were 2.64-fold, 3.56-fold, and 3.26-fold higher than those in the APP/PS1 group, respectively (Fig. 6A). The mRNA levels of NRF1, NRF2, and TFAM in the hippocampus of APP/PS1 mice were significantly reduced.

However, in the metformin and high-dose ZGW group, these mRNA levels notably increased and showed around a two-fold increase compared to the APP/PS1 group (Fig. 6B). Western blot analysis showed that AMPK α , p-AMPK α , and PGC-1 α protein expression was significantly reduced in the APP/PS1 group. In contrast, the expression of these proteins was upregulated in the metformin and high-dose ZGW group (Fig. 6C–F). Consistent with the observed mRNA expression patterns, the decreased protein levels of NRF1 (Fig. 6G) and TFAM (Fig. 6I) in the hippocampus of APP/PS1 mice were restored by metformin and ZGW treatment, although this correction was not observed for NRF2 (Fig. 6H).

7. Improvement in mitochondrial number and function as a result of ZGW treatment

COX activity, which serves as a marker of mitochondrial function, exhibited a notable decrease in the APP/PS1 mice compared to the WT mice. Conversely, both the metformin-treated group and ZGW group displayed significantly elevated COX activity levels compared to the APP/PS1 group (Fig. 7A). In addition, the ATP content in the hippocampus of APP/PS1 mice was significantly lower than that in WT mice. However, metformin and high-dose ZGW administration remarkably increased ATP content, elevating it to 2.11-fold and 2.22-fold higher levels than those in the APP/PS1 group (Fig. 7B). Mitochondrial oxidative stress was further investigated by assessing the major mitochondrial antioxidant defense enzyme superoxide dismutase 2 (SOD2)

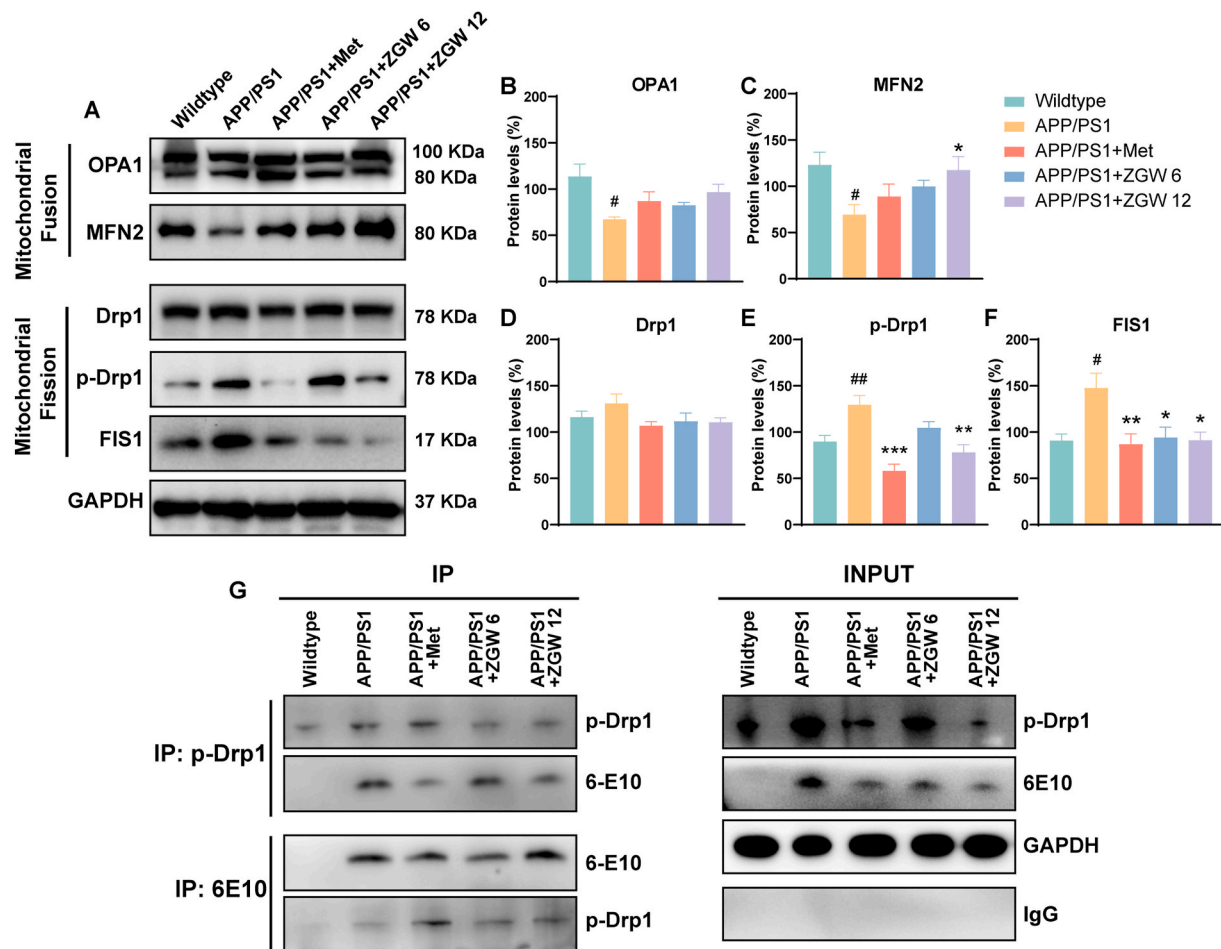


Fig. 5. Inhibition of mitochondrial fragmentation of ZGW via regulation of mitochondrial dynamic (A) Immunoblotting of the expression of OPA1, MFN2, Drp1, p-Drp1, and FIS1. (B–F) Densitometric analysis of the proteins in (A) ($n = 4$). (G) Co-immunoprecipitation coupled with Western blotting confirmed that p-Drp1 interacted with 6E10.

and the oxidative stress product 4-hydroxy-2-nonenal (4-HNE). Western blot analysis showed that treatment with ZGW or metformin effectively mitigated the elevated levels of 4-HNE and concurrently increased SOD2 levels in APP/PS1 mice (Fig. 7C and D). Confocal immunofluorescence studies confirmed that ZGW significantly reduced the accumulation of 4-HNE adducts in the hippocampus of APP/PS1 mice (Fig. 7E).

4. Discussion

Traditional Chinese medicines have shown promise in effectively alleviating cognitive dysfunction in AD (Shi et al., 2017) and may offer a potential new option for the treatment of this disease. In particular, the protective effect of ZGW on cognitive impairment has been reported in streptozotocin-induced AD rat models recently by Liu et al. (2022), who demonstrated that oral administration of 9.46 g/kg ZGW could improve the performance of these AD model mice in the MWM test. However, the effects of ZGW on synaptic loss and synaptic abnormalities, which are strongly correlated with cognitive impairment, remain unknown. The molecular mechanisms related to its effects on mitochondrial function in AD are unclear. In the present study, we provide evidence that oral administration of ZGW for 4 months can improve cognitive function in APP/PS1 mice, as demonstrated by enhanced performance in the MWM test. Moreover, we found that ZGW treatment reduced hippocampal neuron loss and improved synaptic plasticity, as evidenced by increased PSD95 and CtBP2 expression. In addition, ZGW suppressed the increased levels of total A β_{1-42} and A β_{1-40} oligomers in APP/PS1 mice, and it also decreased A β plaques in the hippocampus. Since the accumulation of A β

oligomers is known to result in synaptic loss (Rajmohan and Reddy, 2017), it was considered that the beneficial effect of ZGW on synapses and neurons might be due to its impact on the production and accumulation of A β in APP/PS1 mice.

As a precursor protein to the A β peptide, APP is synthesized in the endoplasmic reticulum and then transported through the Golgi and trans-Golgi network (TGN) to the plasma membrane. Disrupting APP transport to the cell surface and enhancing its internalization will extend the time of APP available for cleavage by β - and γ -secretases within the TGN and endosomes. This process ultimately promotes A β production. Further, increased phosphorylation of APP at threonine 668 is crucial for the subcellular trafficking and binding of APP to cytosolic interactors such as BACE1 (Chang et al., 2006; Triaca et al., 2016), and the result is increased A β production (Allnutt et al., 2020; Haass et al., 2012; Shin et al., 2007; Triaca et al., 2016). Our study supported the notion that the reduction of A β accumulation by ZGW is primarily due to its effect on the trafficking of APP via inhibition of p-APP^{T668} rather than the induction of its degradation. An increasing number of proteins, such as mint (X11) family members, phospholipase D1, and Rab6, are being identified as factors that could interact with APP or act as regulators of APP trafficking (Zhang et al., 2011). Whether these trafficking factors are involved in regulating A β production by ZGW requires further in-depth investigation and validation in future studies.

It is becoming clear that mitochondrial homeostasis dysregulation occurs in neurons prior to A β deposition and NFT formation. This will lead to impaired mitochondrial transport, increased reactive oxygen species, impaired mitochondrial function, and decreased ATP

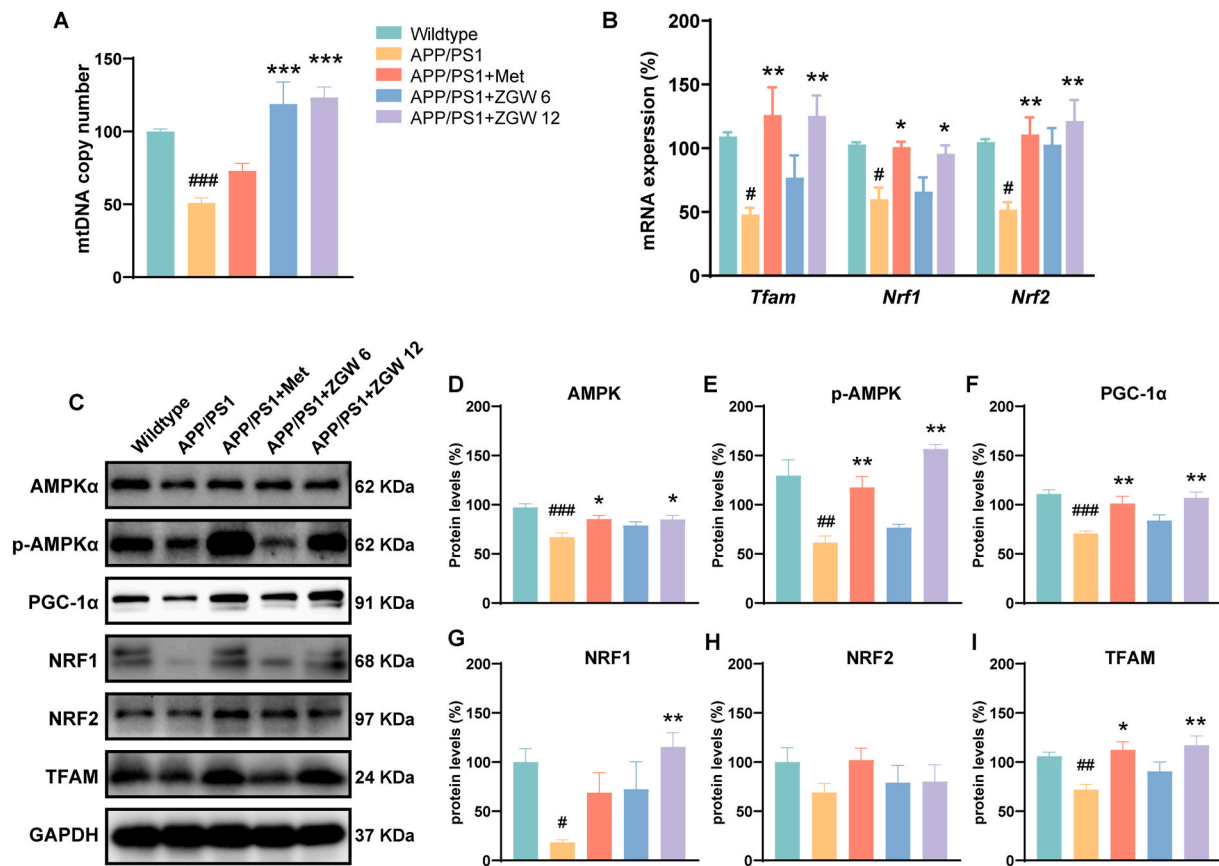


Fig. 6. Promotion of mitochondrial biogenesis by ZGW via regulation of the AMPK/PGC-1 α pathway in the hippocampus (A) qPCR analysis of mtDNA-CN (COX1/18sRNA). (B) qPCR analysis of TFAM, Nrf1, and Nrf2. B2M was applied as an internal reference. (C) Immunoblotting of AMPK, p-AMPK, PGC-1 α , NRF1, NRF2, and TFAM protein expression. (D–I) Densitometric analysis of immunoblotting results of proteins in (C) (n = 4).

production, which collectively promote abnormal mitochondrial fission and accelerate AD progression. T (Han et al., 2023; Xie et al., 2022). In this study, mitochondria from the hippocampus of 10-month-old APP/PS1 mice exhibited significantly greater rates of fission that corresponded to p-Drp1 levels and, in parallel, also exhibited lower rates of fusion based on MFN2 and OPA1 expression. These observations are indicative of altered mitochondrial dynamics in AD mice. Extensive research has shown that Drp1 phosphorylation on specific serine residues, particularly serine 616 (S616), increases Drp1's activity through its translocation from the cytosol to mitochondria, subsequently leading to mitochondria fragmentation (Cho et al., 2013; Qi et al., 2011; Yu et al., 2011). In APP/PS1 mice, elevated Drp1 activity in the hippocampus, associated with increased oxidative stress, was observed as early as 4–5 months of age (Misrani et al., 2021), corresponding to the early stage of AD progression. Based on these previous findings, we speculated whether ZGW may modulate mitochondrial dynamics by regulating Drp1 protein phosphorylation. Our results indicated that ZGW may ameliorate excessive mitochondrial fragmentation by down-regulating Drp1 phosphorylation at S616 and decreasing Drp1 activity and its binding to FIS1 on the mitochondrial membrane. To further confirm the inhibitory effect of ZGW on Drp1, we investigated the interactions between A β and Drp1. It has recently been recognized that A β enters cells and accumulates in mitochondria via a mitochondrial import mechanism (Hansson Petersen et al., 2008). At synaptic terminals, A β has been found to specifically bind with mitochondrial proteins, including Drp1 through VDAC1, the most abundant channel protein expressed on the outer mitochondrial membrane (Du et al., 2008; Manczak et al., 2011). These studies have documented that the co-localization of A β with mitochondria causes structural and functional damage to mitochondria. Further, several researchers have reported the

interaction of A β with Drp1 and the subsequent increase in mitochondrial fragmentation (Kim et al., 2016; Manczak et al., 2011). These studies demonstrated that the enhanced interaction between A β and Drp1 is a key factor in abnormal mitochondrial dynamics, which is the primary cause of synaptic damage. It could be a promising therapeutic strategy to reduce mitochondrial fragmentation. Conversely, modulating this interaction could serve as a promising therapeutic approach to mitigate mitochondrial fragmentation and prevent neuronal and synaptic injury. (Duan et al., 2020; Lin et al., 2022; Tong et al., 2024). Accordingly, in the present study, we used Drp1 immunoprecipitation/immunoblotting analysis of A β antibodies 6E10 to explore whether Drp1 interacts with A β monomers in 10-month-old APP/PS1 mice and found that such abnormal interactions were significantly reduced by treatment with ZGW. ZGW acted as a Drp1 inhibitor, restoring mitochondrial dynamics and improving cognitive decline in APP/PS1 mice.

Mitochondria within the AD milieu exhibit not only an imbalance in A β -related mitochondrial fission but also impaired mitochondrial biogenesis. Deficits in mitochondrial bioenergetics and reduced metabolic activity in the brain, along with increased mitochondrial oxidative stress, are significant pathogenic precursors that may contribute to the onset of Alzheimer's disease (Lin and Beal, 2006; Sheng et al., 2012). PGC-1 α is a key stimulator and sensitive indicator of mitochondrial biogenesis. It activates NRF1, which regulates nuclear genes encoding mitochondrial proteins such as TFAM (Chen et al., 2023a). Activating the PGC-1 α /NRF1/TFAM pathway plays a crucial role in cellular energy metabolism by facilitating the synthesis of mitochondrial DNA and proteins. This activated pathway promotes the biogenesis of new mitochondria, offering a potential strategy to restore mitochondrial bioenergetic deficits in AD. Notably, recent studies indicated that metformin, a well-known pharmacological activator of AMPK that can

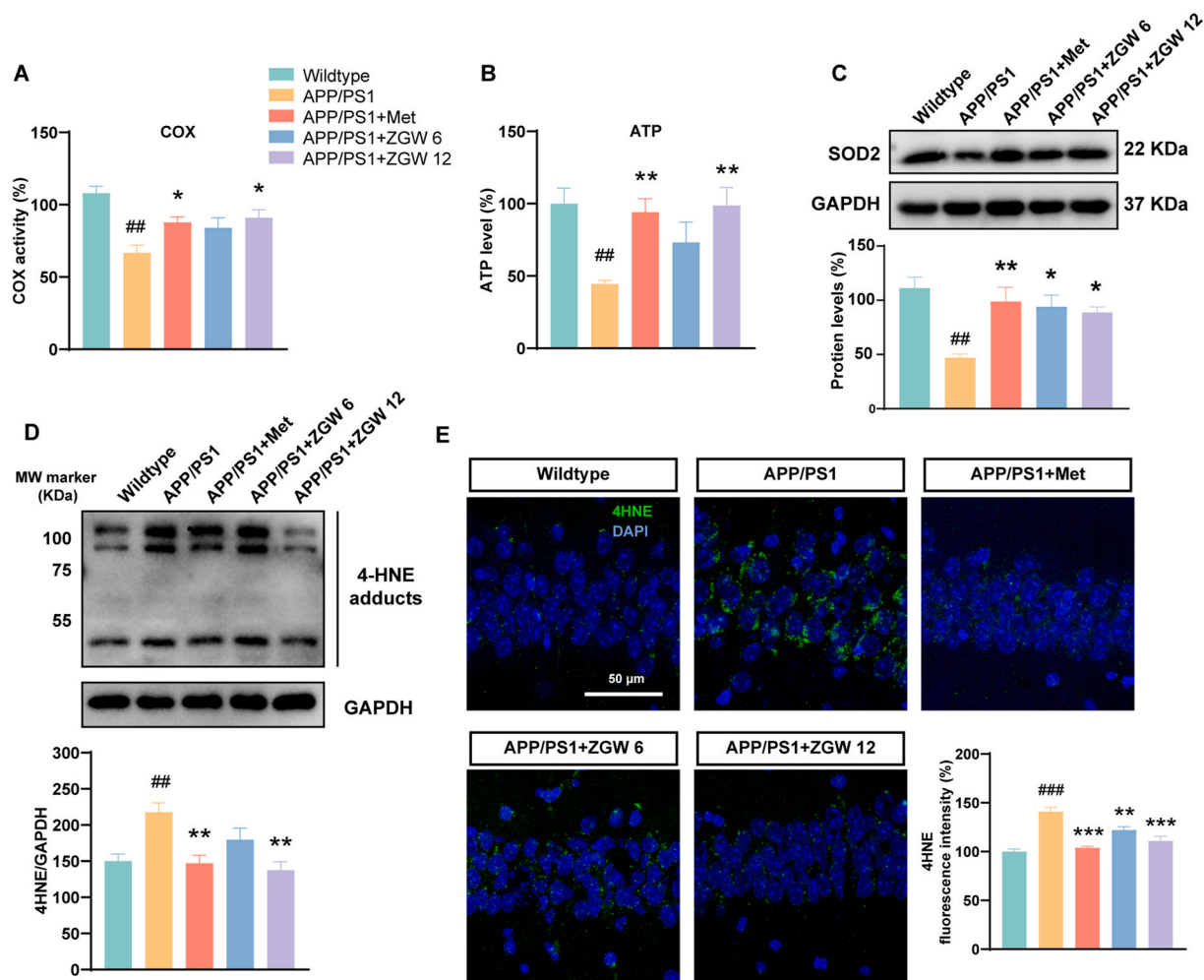


Fig. 7. ZGW-mediated improvement in mitochondrial function

(A) Hippocampal mitochondria were isolated and were evaluated for the Cytochrome c oxidase activities. (B) ATP contents of hippocampal mitochondria were measured using luciferase. (C) Immunoblotting of protein expression of SOD2 and densitometric analysis of the blots (D) Immunoblotting of protein expression of 4-HNE and densitometric analysis of the blots (E) Immunofluorescent staining result of 4-HNE (Green) and DAPI (Blue) expression in the hippocampal neurons and semiquantitative analysis of 4-HNE expression in each group. (n = 5). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

increase the phosphorylation of AMPK α at Thr-172, can beneficially influence mitochondrial dysfunction and significantly improve cognitive deficits in animal models of AD (Meng et al., 2015; Ou et al., 2018; Sanati et al., 2022). In this study, we used metformin as a positive control to evaluate the effect of ZGW on AMPK/PGC-1 α -driven mitochondrial biogenesis. Our data showed that reduced activity of AMPK and PGC-1 α in APP/PS1 mice was rescued by ZGW or metformin. Similar to metformin, ZGW treatment significantly increased the mitochondrial DNA copy number and mRNA level of mitochondrial genes (i. e., Nrf1/2 and TFAM) in the hippocampus of APP/PS1 mice; this implies that the ZGW-induced augmentation of hippocampus mitochondrial biogenesis is related to the AMPK/PGC-1 α pathway. Furthermore, like metformin, ZGW significantly improved mitochondrial function, as indicated by increased mitochondrial DNA copy number, restored COX activity, and elevated ATP production. In addition, ZGW significantly enhanced mitochondrial antioxidant defense by increasing SOD2 expression and reduced the levels of the most stable lipid peroxidation product, 4-HNE adducts, thus preventing neuronal death from oxidative stress. Collectively, our results demonstrate that ZGW not only attenuates excessive mitochondrial division but also enhances mitochondrial biogenesis in the hippocampus of APP/PS1 mice. Moreover, our observations indicate that AMPK might be the key target of ZGW in regulating mitochondrial quality. To further clarify the mechanisms underlying

ZGW's effects on mitochondrial dynamics and biogenesis in AD, future studies should include rescue experiments using specific inhibitors targeting Drp1, PGC-1 α , and NRF1.

In summary, our results in the present AD mouse model demonstrate that ZGW exerts protective effects on AD by addressing multiple pathological processes, including neuron loss, impaired synaptic plasticity, and A β plaque formation. Furthermore, the present study shows that ZGW reduces Drp1-mediated mitochondrial fission induced by A β and activates AMPK/PGC-1 α -regulated mitochondrial bioenergetics in APP/PS1 mice. These findings suggest that ZGW may be a promising therapeutic agent for AD.

CRediT authorship contribution statement

Xirui Yan: Writing – original draft, Methodology, Investigation, Formal analysis. **Yifang Yang:** Investigation, Formal analysis, Data curation. **Weiling Huang:** Investigation, Formal analysis, Data curation. **Shuping Fu:** Resources, Investigation, Data curation. **Bo Cui:** Resources, Investigation, Data curation. **Min Chu:** Resources, Methodology. **Yang Dong:** Resources, Methodology. **Yinting Peng:** Resources, Methodology. **Haiyan Song:** Resources, Methodology. **Jianrong Shi:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Qing Liu:** Writing – review & editing, Supervision, Funding

acquisition, Conceptualization.

Funding

This work was supported by the National Natural Science Foundation of China (Grant No. 82074141), Shanghai Jiao Tong University Trans-med Awards Research (STAR Project, 20230101), and Shanghai Municipal Commission of Health and Family Planning (Grant No. 202340284).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.119425>.

Data availability

Data will be made available on request.

References

- Allnutt, A.B., Waters, A.K., Kesari, S., Yenugonda, V.M., 2020. Physiological and pathological roles of Cdk5: potential directions for therapeutic targeting in neurodegenerative disease. *ACS Chem. Neurosci.* 11, 1218–1230. <https://doi.org/10.1021/acscchemneuro.0c00096>.
- Bobba, A., Amadoro, G., Valenti, D., Corsetti, V., Lassandro, R., Atlante, A., 2013. Mitochondrial respiratory chain Complexes I and IV are impaired by beta-amyloid via direct interaction and through Complex I-dependent ROS production, respectively. *Mitochondrion* 13, 298–311. <https://doi.org/10.1016/j.mito.2013.03.008>.
- Cen, S., Qian, X., Wu, C., Xu, X., Yang, X., 2023. Efficacy and clinical significance of the Zuogui pill on premature ovarian failure via the GDF-9/smad2 pathway. *Nutr. Cancer* 75, 488–497. <https://doi.org/10.1080/01635581.2022.2123531>.
- Chang, K.A., Kim, H.S., Ha, T.Y., Ha, J.W., Shin, K.Y., Jeong, Y.H., Lee, J.P., Park, C.H., Kim, S., Baik, T.K., Suh, Y.H., 2006. Phosphorylation of amyloid precursor protein (APP) at Thr668 regulates the nuclear translocation of the APP intracellular domain and induces neurodegeneration. *Mol. Cell Biol.* 26, 4327–4338. <https://doi.org/10.1128/MCB.02393-05>.
- Chen, M., Yan, R., Luo, J., Ning, J., Zhou, R., Ding, L., 2023a. The role of PGC-1alpha-mediated mitochondrial biogenesis in neurons. *Neurochem. Res.* 48, 2595–2606. <https://doi.org/10.1007/s11064-023-03934-8>.
- Chen, W., Zhao, H., Li, Y., 2023b. Mitochondrial dynamics in health and disease: mechanisms and potential targets. *Signal Transduct. Targeted Ther.* 8, 333. <https://doi.org/10.1038/s41392-023-01547-9>.
- Cho, B., Choi, S.Y., Cho, H.M., Kim, H.J., Sun, W., 2013. Physiological and pathological significance of dynamin-related protein 1 (drp1)-dependent mitochondrial fission in the nervous system. *Exp. Neurobiol.* 22, 149–157. <https://doi.org/10.5607/en.2013.22.3.149>.
- Deng, Z., 2017. *Formulas of Traditional Chinese Medicine (Chinese Edition)*. China Press of Traditional Chinese Medicine, Beijing, China.
- Du, H., Guo, L., Fang, F., Chen, D., Sosunov, A.A., McKhann, G.M., Yan, Y., Wang, C., Zhang, H., Molkentin, J.D., Gunn-Moore, F.J., Vonsattel, J.P., Arancio, O., Chen, J. X., Yan, S.D., 2008. Cyclophilin D deficiency attenuates mitochondrial and neuronal perturbation and ameliorates learning and memory in Alzheimer's disease. *Nat. Med.* 14, 1097–1105. <https://doi.org/10.1038/nm.1868>.
- Duan, C., Wang, L., Zhang, J., Xiang, X., Wu, Y., Zhang, Z., Li, Q., Tian, K., Xue, M., Liu, L., Li, T., 2020. Mdivi-1 attenuates oxidative stress and exerts vascular protection in ischemic/hypoxic injury by a mechanism independent of Drp1 GTPase activity. *Redox Biol.* 37, 101706. <https://doi.org/10.1016/j.redox.2020.101706>.
- Eckert, G.P., Renner, K., Eckert, S.H., Eckmann, J., Hagl, S., Abdel-Kader, R.M., Kurz, C., Leuner, K., Muller, W.E., 2012. Mitochondrial dysfunction—a pharmacological target in Alzheimer's disease. *Mol. Neurobiol.* 46, 136–150. <https://doi.org/10.1007/s12035-012-8271-z>.
- Fang, E.F., Hou, Y., Palikaras, K., Adriaanse, B.A., Kerr, J.S., Yang, B., Lautrup, S., Hasan-Olive, M.M., Caponio, D., Dan, X., Rocktaschel, P., Croteau, D.L., Akbari, M., Greig, N.H., Fladby, T., Nilsen, H., Cader, M.Z., Mattson, M.P., Tavernarakis, N., Bohr, V.A., 2019. Mitophagy inhibits amyloid-beta and tau pathology and reverses cognitive deficits in models of Alzheimer's disease. *Nat. Neurosci.* 22, 401–412. <https://doi.org/10.1038/s41593-018-0332-9>.
- Farr, S.A., Roesler, E., Niehoff, M.L., Roby, D.A., McKee, A., Morley, J.E., 2019. Metformin improves learning and memory in the SAMP8 mouse model of Alzheimer's disease. *J. Alzheimers Dis.* 68, 1699–1710. <https://doi.org/10.3233/JAD-181240>.
- Fu, X., Jiang, X., Chen, X., Zhu, L., Zhang, G., 2019. The differential expression of mitochondrial function-associated proteins and antioxidant enzymes during bovine herpesvirus 1 infection: a potential mechanism for virus infection-induced oxidative mitochondrial dysfunction. *Mediat. Inflamm.* 2019, 7072917. <https://doi.org/10.1155/2019/7072917>.
- Grel, H., Woznica, D., Ratajczak, K., Kalwarczyk, E., Anchimowicz, J., Switlik, W., Olejnik, P., Zielonka, P., Stobiecka, M., Jakiela, S., 2023. Mitochondrial dynamics in neurodegenerative diseases: unraveling the role of fusion and fission processes. *Int. J. Mol. Sci.* 24, 13033. <https://doi.org/10.3390/ijms241713033>.
- Haass, C., Kaether, C., Thinakaran, G., Sisodia, S., 2012. Trafficking and proteolytic processing of APP. *Cold Spring Harb Perspect Med* 2, a006270. <https://doi.org/10.1101/cshperspect.a006270>.
- Han, Y., Liu, D., Cheng, Y., Ji, Q., Liu, M., Zhang, B., Zhou, S., 2023. Maintenance of mitochondrial homeostasis for Alzheimer's disease: strategies and challenges. *Redox Biol.* 63, 102734. <https://doi.org/10.1016/j.redox.2023.102734>.
- Hansson Petersen, C.A., Alikhani, N., Behbahani, H., Wiehager, B., Pavlov, P.F., Alafuzoff, I., Leinonen, V., Ito, A., Winblad, B., Glaser, E., Ankarcrona, M., 2008. The amyloid beta-peptide is imported into mitochondria via the TOM import machinery and localized to mitochondrial cristae. *Proc. Natl. Acad. Sci. U. S. A.* 105, 13145–13150. <https://doi.org/10.1073/pnas.0806192105>.
- Islam, B.U., Jabir, N.R., Tabrez, S., 2019. The role of mitochondrial defects and oxidative stress in Alzheimer's disease. *J. Drug Target.* 27, 932–942. <https://doi.org/10.1080/1061186X.2019.1584808>.
- Jia, K., Tian, J., Wang, T., Guo, L., Xuan, Z., Swerdlow, R.H., Du, H., 2023. Mitochondria-sequestered Abeta renders synaptic mitochondria vulnerable in the elderly with a risk of Alzheimer disease. *JCI Insight* 8, e174290. <https://doi.org/10.1172/jci.insight.174290>.
- Kim, D.I., Lee, K.H., Gabr, A.A., Choi, G.E., Kim, J.S., Ko, S.H., Han, H.J., 2016. Abeta-Induced Drp1 phosphorylation through Akt activation promotes excessive mitochondrial fission leading to neuronal apoptosis. *Biochim. Biophys. Acta* 1863, 2820–2834. <https://doi.org/10.1016/j.bbamer.2016.09.003>.
- Li, N., Yan, X., Huang, W., Chu, M., Dong, Y., Song, H., Peng, Y., Shi, J., Liu, Q., 2023. Curcumin protects against the age-related hearing loss by attenuating apoptosis and senescence via activating Nrf2 signaling in cochlear hair cells. *Biochem. Pharmacol.* 212, 115575. <https://doi.org/10.1016/j.bcp.2023.115575>.
- Li, W., Liu, Z., Liu, L., Yang, F., Li, W., Zhang, K., Liu, Z., Cheng, Y., Yin, J., Sun, Y., 2018. Effect of Zuogui pill and Yougui pill on osteoporosis: a randomized controlled trial. *J. Tradit. Chin. Med.* 38, 33–42.
- Li, X., Shi, Q., Xu, H., Xiong, Y., Wang, C., Le, L., Lian, J., Wu, G., Peng, F., Liu, Q., Du, X., 2022. Ebselen interferes with Alzheimer's disease by regulating mitochondrial function. *Antioxidants* 11, 1350. <https://doi.org/10.3390/antiox11071350>.
- Lin, M.T., Beal, M.F., 2006. Mitochondrial dysfunction and oxidative stress in neurodegenerative diseases. *Nature* 443, 787–795. <https://doi.org/10.1038/nature05292>.
- Lin, N., Jin, J.W., Lai, Z.M., Zhang, D.F., Chen, Y., Guo, H.G., Liu, J.L., 2022. Mdivi-1 improves postoperative neurocognitive disorders in aged rats undergoing splenectomy by inhibiting dynamin-related protein-1. *Ann. Transl. Med.* 10, 1338. <https://doi.org/10.21037/atm-22-5496>.
- Liu, F., Tan, F., Tong, W., Fan, Q., Ye, S., Lu, S., Teng, Z., Han, M., Zhang, M., Chai, Y., 2018. Effect of Zuoguiwan on osteoporosis in ovariectomized rats through RANKL/OPG pathway mediated by beta2AR. *Biomed. Pharmacother.* 103, 1052–1060. <https://doi.org/10.1016/j.biopha.2018.04.102>.
- Liu, H., Zhong, L., Dai, Q., Yang, J., Zhang, Y., Zhang, B., Jiang, Y., 2022. Zuoguiwan ameliorates cognitive deficits and neuro-inflammation in streptozotocin-induced Alzheimer's disease rats. *Neuroimmunomodulation* 29, 63–69. <https://doi.org/10.1159/000516396>.
- Manczak, M., Anekonda, T.S., Henson, E., Park, B.S., Quinn, J., Reddy, P.H., 2006. Mitochondria are a direct site of A beta accumulation in Alzheimer's disease neurons: implications for free radical generation and oxidative damage in disease progression. *Hum. Mol. Genet.* 15, 1437–1449. <https://doi.org/10.1093/hmg/ddl066>.
- Manczak, M., Calkins, M.J., Reddy, P.H., 2011. Impaired mitochondrial dynamics and abnormal interaction of amyloid beta with mitochondrial protein Drp1 in neurons from patients with Alzheimer's disease: implications for neuronal damage. *Hum. Mol. Genet.* 20, 2495–2509. <https://doi.org/10.1093/hmg/ddr139>.
- Meng, S., Cao, J., He, Q., Xiong, L., Chang, E., Radovick, S., Wondisford, F.E., He, L., 2015. Metformin activates AMP-activated protein kinase by promoting formation of the alphabeta gamma heterotrimeric complex. *J. Biol. Chem.* 290, 3793–3802. <https://doi.org/10.1074/jbc.M114.604421>.
- Mishra, E., Thakur, M.K., 2023. Mdivi-1 rescues memory decline in scopolamine-induced amnesic male mice by ameliorating mitochondrial dynamics and hippocampal plasticity. *Mol. Neurobiol.* 60, 5426–5449. <https://doi.org/10.1007/s12035-023-03397-6>.
- Misrani, A., Tabassum, S., Huo, Q., Tabassum, S., Jiang, J., Ahmed, A., Chen, X., Zhou, J., Zhang, J., Liu, S., Feng, X., Long, C., Yang, L., 2021. Mitochondrial deficits with neural and social damage in early-stage Alzheimer's disease model mice. *Front. Aging Neurosci.* 13, 748388. <https://doi.org/10.3389/fnagi.2021.748388>.
- Nair, A.B., Jacob, S., 2016. A simple practice guide for dose conversion between animals and human. *J. Basic Clin. Pharm.* 7, 27–31. <https://doi.org/10.4103/0976-0105.177703>.
- Niedowicz, D.M., Ozcan, S., Nelson, P.T., 2018. Glimepiride administered in chow reversibly impairs glucose tolerance in mice. *J. Diabetes Res.* 1251345. <https://doi.org/10.1155/2018/1251345>.

- Oka, S., Leon, J., Sakumi, K., Ide, T., Kang, D., LaFerla, F.M., Nakabeppu, Y., 2016. Human mitochondrial transcriptional factor A breaks the mitochondria-mediated vicious cycle in Alzheimer's disease. *Sci. Rep.* 6, 37889. <https://doi.org/10.1038/srep37889>.
- Ou, Z., Kong, X., Sun, X., He, X., Zhang, L., Gong, Z., Huang, J., Xu, B., Long, D., Li, J., Li, Q., Xu, L., Xuan, A., 2018. Metformin treatment prevents amyloid plaque deposition and memory impairment in APP/PS1 mice. *Brain Behav. Immun.* 69, 351–363. <https://doi.org/10.1016/j.bbi.2017.12.009>.
- Qi, X., Disatnik, M.H., Shen, N., Sobel, R.A., Mochly-Rosen, D., 2011. Aberrant mitochondrial fission in neurons induced by protein kinase Cdelta under oxidative stress conditions in vivo. *Mol. Biol. Cell* 22, 256–265. <https://doi.org/10.1091/mbc.E10-06-0551>.
- Rajmohan, R., Reddy, P.H., 2017. Amyloid-beta and phosphorylated tau accumulations cause abnormalities at synapses of alzheimer's disease neurons. *J Alzheimers Dis* 57, 975–999. <https://doi.org/10.3233/JAD-160612>.
- Sanati, M., Aminyavari, S., Afshari, A.R., Sahebkar, A., 2022. Mechanistic insight into the role of metformin in Alzheimer's disease. *Life Sci.* 291, 120299. <https://doi.org/10.1016/j.lfs.2021.120299>.
- Sheng, B., Wang, X., Su, B., Lee, H.G., Casadesus, G., Perry, G., Zhu, X., 2012. Impaired mitochondrial biogenesis contributes to mitochondrial dysfunction in Alzheimer's disease. *J. Neurochem.* 120, 419–429. <https://doi.org/10.1111/j.1471-4159.2011.07581.x>.
- Shi, J., Ni, J., Lu, T., Zhang, X., Wei, M., Li, T., Liu, W., Wang, Y., Shi, Y., Tian, J., 2017. Adding Chinese herbal medicine to conventional therapy brings cognitive benefits to patients with Alzheimer's disease: a retrospective analysis. *BMC Compl. Alternative Med.* 17, 533. <https://doi.org/10.1186/s12906-017-2040-5>.
- Shin, R.W., Ogino, K., Shimabuku, A., Taki, T., Nakashima, H., Ishihara, T., Kitamoto, T., 2007. Amyloid precursor protein cytoplasmic domain with phospho-Thr668 accumulates in Alzheimer's disease and its transgenic models: a role to mediate interaction of Abeta and tau. *Acta Neuropathol.* 113, 627–636. <https://doi.org/10.1007/s00401-007-0211-z>.
- Thomas, R.R., Khan, S.M., Smigrodzki, R.M., Onyango, I.G., Dennis, J., Khan, O.M., Portelli, F.R., Bennett Jr., J.P., 2012. RbTFAM treatment stimulates mitochondrial oxidative metabolism and improves memory in aged mice. *Aging (Albany NY)* 4, 620–635. <https://doi.org/10.18632/aging.100488>.
- Tong, Z., Du, X., Zhou, Y., Jing, F., Ma, J., Feng, Y., Lou, S., Wang, Q., Dong, Z., 2024. Drp1-mediated mitochondrial fission promotes pulmonary fibrosis progression through the regulation of lipid metabolic reprogramming by ROS/HIF-1alpha. *Cell. Signal.* 117, 111075. <https://doi.org/10.1016/j.cellsig.2024.111075>.
- Triaca, V., Sposato, V., Bolasco, G., Ciotti, M.T., Pellicci, P., Bruni, A.C., Cupidi, C., Maletta, R., Feligioni, M., Nistico, R., Canu, N., Calissano, P., 2016. NGF controls APP cleavage by downregulating APP phosphorylation at Thr668: relevance for Alzheimer's disease. *Aging Cell* 15, 661–672. <https://doi.org/10.1111/ace.12473>.
- Wang, W., Yin, J., Ma, X., Zhao, F., Siedlak, S.L., Wang, Z., Torres, S., Fujioka, H., Xu, Y., Perry, G., Zhu, X., 2017. Inhibition of mitochondrial fragmentation protects against Alzheimer's disease in rodent model. *Hum. Mol. Genet.* 26, 4118–4131. <https://doi.org/10.1093/hmg/ddx299>.
- Wang, W., Zhao, F., Lu, Y., Siedlak, S.L., Fujioka, H., Feng, H., Perry, G., Zhu, X., 2023. Damaged mitochondria coincide with presynaptic vesicle loss and abnormalities in alzheimer's disease brain. *Acta Neuropathol Commun* 11, 54. <https://doi.org/10.1186/s40478-023-01552-7>.
- Wang, W., Zhao, F., Ma, X., Perry, G., Zhu, X., 2020. Mitochondria dysfunction in the pathogenesis of Alzheimer's disease: recent advances. *Mol. Neurodegener.* 15, 1–22. <https://doi.org/10.1186/s13024-020-00376-6>.
- Xie, W., Guo, D., Li, J., Yue, L., Kang, Q., Chen, G., Zhou, T., Wang, H., Zhuang, K., Leng, L., Li, H., Chen, Z., Gao, W., Zhang, J., 2022. CEND1 deficiency induces mitochondrial dysfunction and cognitive impairment in Alzheimer's disease. *Cell Death Differ.* 29, 2417–2428. <https://doi.org/10.1038/s41418-022-01027-7>.
- Yakushiji, Y., Kawamoto, K., Uchihashi, K., Ihara, M., Aoki, S., Nagaishi, Y., Suzuyama, K., Tsugitomi, Y., Hara, H., 2020. Low-dose phosphodiesterase III inhibitor reduces the vascular amyloid burden in amyloid-beta protein precursor transgenic mice. *Int. J. Mol. Sci.* 21. <https://doi.org/10.3390/ijms21072295>.
- Yao, J.P., Li, Y.M., Feng, Y.M., Ma, S.J., Long, X.Y., 2017. Effect of bushen formula on content of serum CORT of natural aging rat and ultrastructure of hippocampal dentate gyrus. *New Chinese Medicine* 49, 1–4. <https://doi.org/10.13457/j.cnki.jncm.2017.06.001>. Chinese.
- Yu, T., Jhun, B.S., Yoon, Y., 2011. High-glucose stimulation increases reactive oxygen species production through the calcium and mitogen-activated protein kinase-mediated activation of mitochondrial fission. *Antioxidants Redox Signal.* 14, 425–437. <https://doi.org/10.1089/ars.2010.3284>.
- Yu, T., Xing, Y., Gao, Q., Wang, D., Chen, H., Wang, H., Zhang, Y., 2023. Ginkgo biloba extract drives gut flora and microbial metabolism variation in a mouse model of alzheimer's disease. *Pharmaceutics* 15. <https://doi.org/10.3390/pharmaceutics15122746>.
- Zhang, Y.W., Thompson, R., Zhang, H., Xu, H., 2011. APP processing in Alzheimer's disease. *Mol. Brain* 4, 3. <https://doi.org/10.1186/1756-6606-4-3>.
- Zhao, X.Y., Li, X.T., Zhou, F., 2019. Zuoguiwan improves the learning ability of aging rats by neuron's mitochondria pathway. *Chinese Journal of Gerontology* 39, 3722–3725. Chinese.
- Zysk, M., Beretta, C., Naia, L., Dakhel, A., Pavenius, L., Brismar, H., Lindskog, M., Ankarcrona, M., Erlandsson, A., 2023. Amyloid-beta accumulation in human astrocytes induces mitochondrial disruption and changed energy metabolism. *J. Neuroinflammation* 20, 43. <https://doi.org/10.1186/s12974-023-02722-z>.